

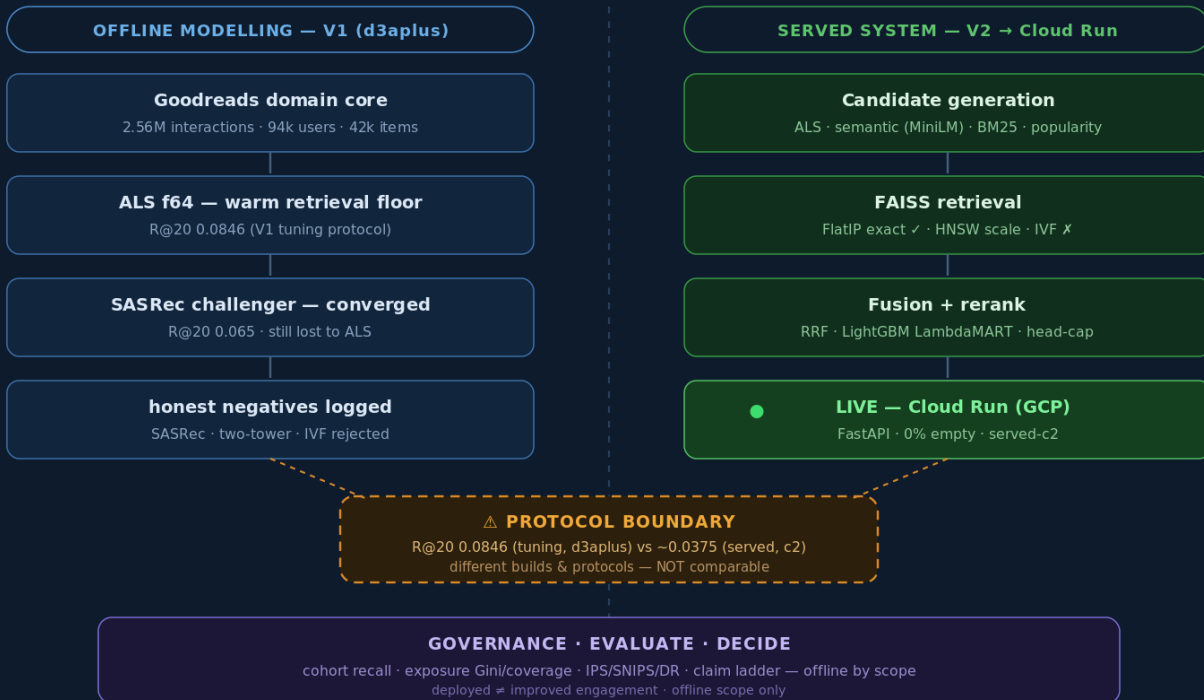
PulseDiscover

Interview Defense Dossier

Offline, honesty-gated recommender **decision** system — 2.56M Goodreads interactions

Python	FastAPI	FAISS	LambdaMART	OPE: IPS/SNIPS/DR	Cloud Run: LIVE
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PULSEDISCOVER — OFFLINE MODELLING vs SERVED SYSTEM



gated program G11-G31 · every stage adopted or rejected with an artifact in outputs/evidence/

STATUS

V1 **gold_candidate 8.9** (locked, terminal) · V2 lane **offline gold-complete** (RiskFrame 9.1) · serving API **deployed to Google Cloud Run** (live HTTPS) · V3 gate **two-tower + Thompson exploration** (offline, honest negative + explore/exploit tradeoff).

READING RULE

Every number in this dossier is **offline and protocol-tagged**. The V1 d3aplus result (R@20 0.0846) and the served-c2 result (R@20 ~0.0375) are **different builds and protocols — NOT comparable**. No online-lift, real-user, or production-quality claim appears anywhere.

Contents

#	Section
1	Project summary & thesis
2	System architecture
3	Fundamentals bridge — 14 method cards
4	Decision log
5	Failures I'm proud of (8 post-mortems)
6	Eval results & model/protocol registry
7	Gate-by-gate walkthrough (G22-G31)
8	Off-policy evaluation deep dive
9	Cold-start & exposure deep dive
10	Truth boundary & claim ladder
11	Product & business reasoning
12	Hard Q&A (22 questions)
13	What I'd build next
14	Evidence artifact index
V3	Neural two-tower + Thompson exploration bandit (G32-G33)

1 Project Summary & Thesis

Thesis. Recommenders rarely fail because the model was wrong. They fail because a metric improved while catalog health, candidate fidelity, or cold-start coverage silently collapsed — and nobody measured the gap. PulseDiscover is the measurement-and-decision layer that catches that gap before it reaches users.

What it is. An offline, honesty-gated two-stage recommender decision system on a real 2.56M-interaction Goodreads fantasy/paranormal corpus (94k users, ~42k items; 40,541 in the served ALS catalog). Four candidate sources (ALS, dense semantic, BM25, popularity) feed a FAISS retrieval layer, a learned ALS+semantic fusion ranker, a config-flagged exposure rerank, and a graceful fallback tree, served by a FastAPI app deployed on Cloud Run — with cohort, exposure, and off-policy evaluation governing every ship decision.

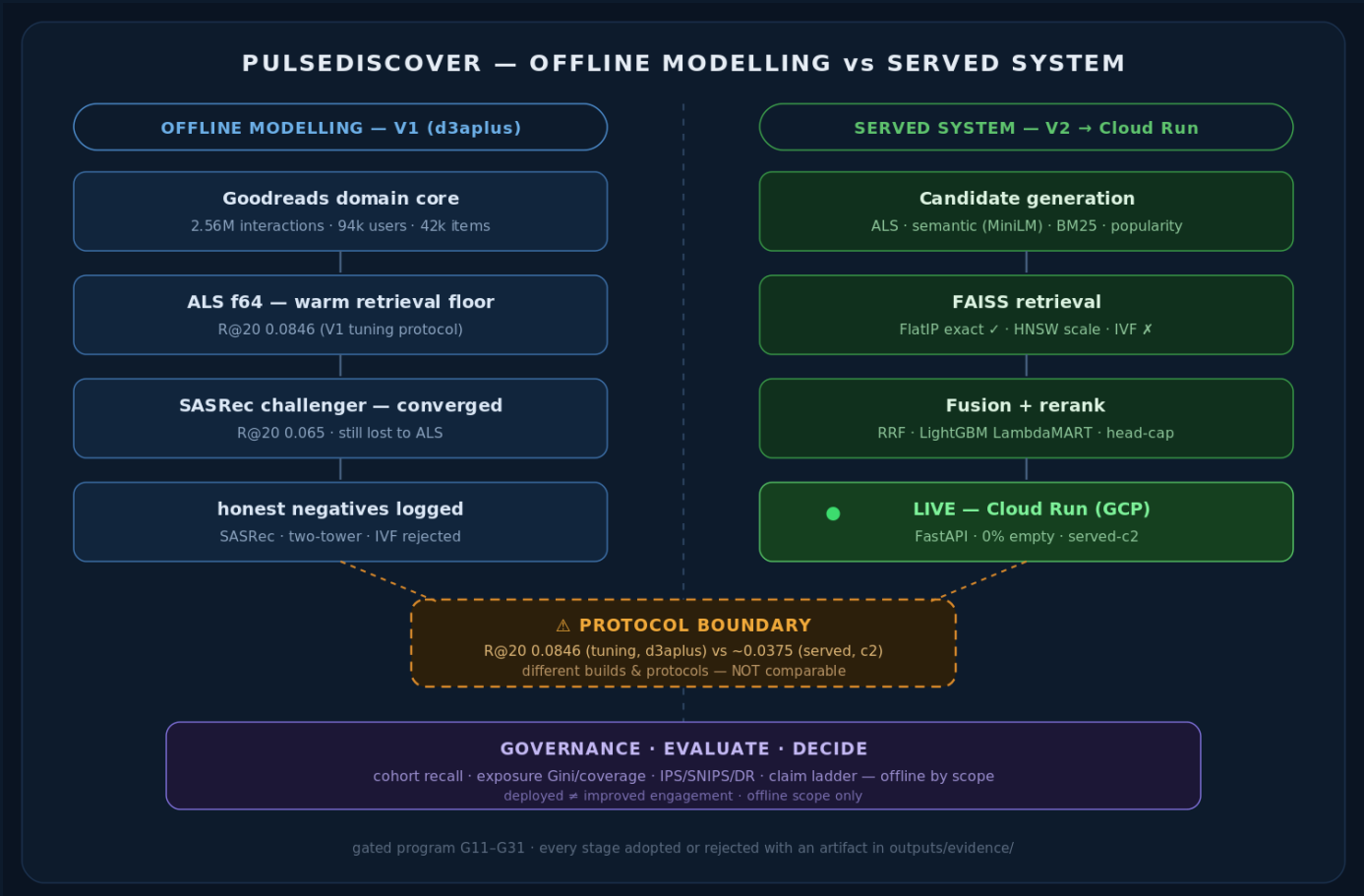
How it was built. As a gated experimentation program (G11-G31). Each stage was added, measured, and either adopted or rejected with an artifact in `outputs/evidence/`. V1 established the offline modelling floor; the V2 lane (G22-G31) added serving, cold-start/fallback quality, exposure governance, a heuristic exposure rerank, dense semantic retrieval, a learned fusion tournament, executed off-policy evaluation, and a search/IR front end — then the serving path shipped to Cloud Run.

Why V1 is gold_candidate, not gold_complete. The canonical SASRec convergence headline is provenance-tagged `plot_reconstructed` (the direct artifact was lost with a Colab session). The decision it supports (MF beats a converged SASRec) is artifact-backed; the packaging is one artifact short of clean — so the status is honestly held at candidate.

What it deliberately is NOT. Not an online/production-quality system, not solved cold-start, not fairness-certified, not an LLM recommender. Those boundaries are load-bearing and stated throughout.

2 System Architecture

Two lanes share one honest boundary. The **offline modelling lane** (V1, d3aplus) produced the findings — ALS is the warm floor, a converged SASRec still lost. The **served system lane** (V2, c2 → Cloud Run) is what actually runs. The amber bridge is the protocol boundary that keeps the two from being conflated.



Per-stage contract

Stage	Input → Output	Claim enabled / not enabled
Candidate gen	interactions → candidate pool	"coverage is the binding lever (+~44% recall)" / not online impact
FAISS retrieval	embeddings → top-K	"FlatIP lossless ~47% faster; HNSW near-lossless" / not quality gain
Fusion / rank	pool+features → ordered slate	"learned fusion beat ALS-only on test" / not in production
Exposure rerank	slate → reranked slate	"interpretable exposure control" / not learned LTR
Fallback	any request → non-empty slate	"0% empty across failure modes" / not fallback quality
Serving	user_id,k,mode → JSON	"production-like, deployed to Cloud Run" / not online lift
Evaluation	logs/splits → metrics	"cohort + exposure + OPE" / not real-traffic value
Decision	frontiers → ship/hold/reject	"I decide what ships and what must not" / not deployed-and-proven

3 Fundamentals Bridge — method cards

Each technique from first principles: objective · mechanism · assumptions · failure modes · why-over-alternatives · role in PulseDiscover.

ALS — implicit-feedback matrix factorization

Objective: $\min \sum c_{ui} (p_{ui} - x_u \cdot y_i)^2 + \lambda (\|x\|^2 + \|y\|^2)$, confidence $c=1+\alpha r$, preference $p=1[r>0]$. **Mechanism:** alternate closed-form ridge solves per side — $x_u = (Y^T C_u Y + \lambda I)^{-1} Y^T C_u p$. **Assumes:** low-rank structure; missing $\approx$ weak-negative. **Fails on:** cold users / niche tail. **Over BPR:** closed-form, fast, stable. **Role:** the warm-retrieval floor and the served model (c2).

SASRec — self-attentive sequential recommendation

Objective: maximize next-item likelihood (full softmax over the catalog). **Mechanism:** causal self-attention over the item-embedding sequence. **Assumes:** informative order + dense-enough sequences. **Fails on:** short/sparse/padded histories (the NaN-mask bug). **Lever found:** full-softmax vs sampled-softmax ($\sim 6.7\times$) — still $<$ ALS. **Role:** the deep-model challenger, trained to convergence; a documented honest negative.

Two-tower — ID + content retrieval (V3, G32)

Objective: user tower = mean-pooled history item-embeddings; item tower = learned-ID embedding $\oplus$ a projection of the 384-d MiniLM content vector; dot-product scored, BPR loss (8 uniform negatives), 23 epochs, PyTorch CPU. **Result:** R@20 **0.0642** [0.057–0.071] on the exact ALS-floor protocol — a documented honest negative vs ALS 0.0846 (non-overlapping CIs). Content ablation: turning the content tower off drops R@20 to 0.0432, so MiniLM content contributes **+0.021 R@20**. **Reading:** content is now first-class inside retrieval and earns a real lift, but still loses to a tuned MF on dense warm book histories — the second neural model to do so (after SASRec).

LightGCN — graph CF (deferred, documented)

Mechanism: light graph convolution (no feature transforms/nonlinearities) propagating CF signal over the user-item graph. **Status:** deferred — graph training over $\sim 42k$ items is not cheap on CPU, and on a catalog where a converged SASRec and a content two-tower both lost to MF it is unlikely to pay. Documented, not silently skipped.

LambdaMART — learning-to-rank fusion

Objective: boosted trees optimizing a listwise NDCG surrogate; gradients are λ -weighted by the NDCG change of swapping a pair, grouped per query (user). **Assumes:** enough positive labels/query; features carry cross-source signal. **Fails on:** extreme label sparsity (overfit). **Role:** the G28 fusion ranker over ALS+semantic+popularity candidate features — champion of the tournament.

FAISS — ANN retrieval (FlatIP / HNSW / IVF)

FlatIP: exact exhaustive inner product ($O(N)$, lossless). **HNSW:** navigable-small-world graph, efSearch trades recall/latency. **IVF:** coarse quantiser, nprobe controls cells scanned. **Failure:** aggressive nprobe shreds the candidate pool while a single-gold metric stays flat. **Role:** FlatIP default, HNSW scale option, IVF rejected.

Dense semantic content retrieval

Objective: embed item content (title + genre shelves) with MiniLM (384-d, L2-normalized); FAISS IndexFlatIP (cosine-equiv); item-to-item from the seed item. **Assumes:** content similarity $\approx$ taste at genre level; metadata text informative (100% coverage). **Fails on:** head relevance (collaborative signal dominates), zero-history users. **Role:** the only source that structurally reaches item-cold-start items (reachability 54.6%).

RRF — reciprocal rank fusion

Objective: $\text{score}(i) = \sum_s 1/(c + \text{rank}_s(i))$ over sources; score-scale-free, training-free. **Assumes:** rank is a comparable cross-source signal. **Fails on:** nothing catastrophic — but it buys robustness, not a recall lift. **Role:** best non-learned fusion; ALS-first fusion fails (saturates top-20), RRF lets semantic compete.

IPS — inverse propensity scoring

Objective: $V(\pi_e) = (1/N) \sum [1\{a_i = \pi_e(x_i)\} / \pi_{\log}(a_i | x_i)] \cdot r_i$. **Assumes:** overlap ($\pi_{\log} > 0$ on π_e 's support); correct propensities. **Fails on:** high variance at modest ESS (here it over-estimated $\sim 2\times$). **Role:** executed in G29; the cautionary high-variance baseline.

SNIPS — self-normalized IPS

Objective: $\sum w_i r_i / \sum w_i$, $w_i = 1\{a_i = \pi_e\} / \pi_{\log}$. Trades a little bias for much lower variance. **Role:** the recommended estimator when no reward model exists — stable and close to the known value in G29.

Doubly-Robust (DR) OPE

Objective: $\text{DR} = \text{mean}(\hat{r}(x, \pi_e)) + \text{mean}(w \cdot (r - \hat{r}(x, a)))$; unbiased if either propensities or reward model is right. **Fails on:** a mis-calibrated reward model (a class-balanced $\hat{r}$ broke it — went negative). **Role:** lowest-bias estimator once $\hat{r}$ is base-rate-calibrated (DR 0.0089 vs true 0.0106).

Position bias & PAL

Problem: top-ranked items get examined more, so click-recall measures position as much as relevance. PAL/IPS de-bias examination from relevance. **Role:** PulseDiscover does position-bias-aware *evaluation* (a two-lane OPE diagnostic) — not an in-ranker correction (that is labelled future work).

Exposure governance — Gini / Lorenz / base-rate lift

Objective: over the full catalog (zeros included) compute $\text{Gini} = 2\sum i \cdot x_i / (n \sum x_i) - (n+1)/n$, the Lorenz curve, top-x% share, and tier exposure-lift vs catalog base rate. **Failure:** Gini alone hides which tail — pair with tier-lift. **Role:** catalog-health as a first-class ship-gate metric (all policies $\text{Gini} > 0.97$; popularity 0.9995).

Cold-start cohorts & fallback policy

Mechanism: cohort users (unknown / sparse 1-2 / low 3-5 / warm 6+); evaluate popularity/content/item-sim/blended fallback; report warm-ALS and cold-fallback metrics SEPARATELY. **Honest nuance:** per-cohort recall is non-monotonic (cold gold is more popularity-predictable) — recall understates the real cost, which is coverage/personalization collapse.

Negative sampling

Idea: sampled negatives approximate the full softmax; popularity-biased samplers skew the gradient. **Role:** implicit in ALS/sequence training; the full-softmax-vs-sampled lever in G20 was exactly this choice.

4 Decision Log

Decision	Rationale	Boundary kept
ALS over SASRec (warm floor)	Converged full-softmax SASRec still lost (0.065 vs 0.085); depth isn't the lever on sparse book histories.	"a converged deep model lost" — not "deep is useless"
FlatIP exact as default	Lossless & ~47% faster; HNSW near-lossless but approximate. Correctness by default, opt into approximation.	HNSW is a flagged scale mode, not silent default
IVF rejected	Low-nprobe kept gold recall flat while candidate-overlap collapsed 0.28–0.78 — a broken pool.	overlap@K is a first-class acceptance check
RRF = robustness not win	Recovers cold-start + ~4× coverage at ~95% of ALS relevance; does NOT beat ALS-only on warm recall.	no "hybrid beats both lanes" claim
Semantic as complement	Uniquely reaches item-cold-start (54.6%) but loses head relevance (0.010 vs 0.087).	cold-start/tail source, not a warm-path replacement
BM25 = query-by-document	No free-text queries exist; the seed item's text is the query (more-like-this).	seed-item, not a free-text search engine
Learned fusion, bounded	LambdaMART beat ALS-only on held-out test (0.036 vs 0.022) but on 81 positives.	claimed on the offline test split only
OPE executed, not faked	Estimators recover a known value offline; props synthetic, reward a proxy.	methodology demonstrated, not a real value
V1/V2 registry	d3aplus 0.0846 and c2 0.0375 are different builds/protocols.	never conflated — kept in a registry
Two-tower = honest negative (V3)	ID+content BPR two-tower R@20 0.064 vs ALS 0.085 (non-overlapping CIs); content ablation adds +0.021.	content earns a lift; MF still wins warm
Bandit = exploration, not relevance (V3)	Thompson raised coverage 17→51% and cut Gini 0.97→0.81 for a ~1.3pp relevance cost.	explore/exploit stated as a COST, offline only

5 Failures I'm Proud Of

The interview value of a project is the failures you catch yourself. Each: what happened → what it revealed → what I did.

SASRec false convergence (NaN val-loss patience)

Training early-stopped at ~6 epochs on an all-NaN val-loss curve (2-block masked-attention NaN on short/padded sequences); a NaN comparison silently satisfied patience while train loss + eval-R@20 were still rising. **Revealed:** "early stop fired" ≠ "converged." **Did:** rejected as invalid, rebuilt the monitor on an eval-R@20 plateau (valid convergence at ~95 epochs).

v2 eval leakage (same set for stopping + final eval)

SASRec v2 used the 5k eval set for BOTH the early-stopping criterion and the final metric — model selection peeking at test. **Revealed:** reusing a held-out set for selection leaks. **Did:** split a separate validation-monitor set; fixed the epoch cap to an exact value.

Stale-script contamination on Colab

Twice, Colab ran a cached v1 script instead of v3 and reported old numbers. **Revealed:** "I updated the script" ≠ "the run used it." **Did:** hardened the notebook to write/run a distinctly-named file with assert guards; provenance-tagged the unrecoverable headline plot_reconstructed.

IVF — the metric can lie

IVF looked fine on gold Recall@20 while candidate-overlap@K vs exact collapsed to 0.28-0.78 — a mostly different candidate set that happened to contain the gold. **Revealed:** a single-gold metric is blind to most of the slate changing. **Did:** rejected IVF; made overlap@K a first-class check.

ID-dtype silent zeroing

book_id is a string in the ALS index but ints in eval; every membership test compared int vs str keys, so every lookup missed and recall floored to ~0 — while the pipeline ran clean with plausible numbers. **Revealed:** silent type coercion is where pipelines rot. **Did:** caught it via an overlap sanity check (0 catalog-overlapping keys = impossible if types matched); normalized to str repo-wide.

G24 recall non-monotonicity

Cold cohorts scored HIGHER recall than warm — the opposite of the expected story. **Revealed:** cold users' next item is more popularity-predictable, so recall understates cold-start cost. **Did:** refused the expected narrative; reframed the real cost as coverage (~59×) + personalization (0.68→0.001) collapse.

G29 Doubly-Robust went negative

DR returned ~-0.4 (impossible for a [0,1] reward). **Revealed:** a class-balanced reward model calibrated to ~0.5 vs a true ~1% base rate broke the DR baseline. **Did:** caught it from the impossible sign; switched to a base-rate-calibrated model; DR then recovered the known value best of the three.

Serving load-path false confidence (caught in the stress test)

The startup dim-assert was tautological (index built from Y, so it can't fail) and a corrupt model pickle returned 0 items (popularity not yet loaded). **Revealed:** a health check that can't fail is worse than none. **Did:** added a real X/Y dim assert; load popularity independently so a degraded service still serves a non-empty fallback.

V3 OPE overlap collapse for the exploration policy

Reusing the G29 IPS/SNIPS/DR estimators on the Thompson bandit, DR went slightly negative and IPS/SNIPS hit 0 — while they cleanly recovered the greedy policy (DR 0.005 vs true 0.006). **Revealed:** the exploration policy picks items the logging policy rarely logs, so effective sample size collapsed 168→21 and the estimators lost all power. **Did:** reported it as the finding, not a bug — you cannot off-policy-evaluate an exploration policy a logging policy doesn't cover; real logged propensities are the prerequisite.

6 Eval Results & Model/Protocol Registry

PROTOCOL WARNING (load-bearing)

V1 **d3aplust** R@20 **0.0846** and V2 **served-c2** R@20 **~0.0375** are different model builds and different evaluation protocols (V2 = full-catalog single-held-out-gold). **They are NOT comparable.** V2 numbers are used for relative cross-policy/cohort structure, never as a headline that overturns V1.

Headline metrics

Result	Number	Protocol tag
ALS warm floor	R@20 0.0846	V1 d3aplust tuning — REAL_OFFLINE
SASRec converged	R@20 0.065 (lost)	V1 — PLOT_RECONSTRUCTED
Served ALS exact	R@20 ~0.0375	V2 served-c2 — NOT_COMPARABLE
FAISS FlatIP	lossless, ~47% lower p95	V2 latency
HNSW ef64	overlap 0.992, ~2.4×	V2 latency
Cold-start fallback	coverage ~59×; pers. 0.68→0.001	V2 served-c2
Semantic reach	54.6% unseen golds	V2 served-c2
Learned fusion	test 0.036 vs 0.022	V2 — 81 positives, directional
OPE	DR 0.0089 vs true 0.0106	V2 — offline, proxy reward
Exposure	Gini >0.97; pop 0.9995	V2 — concentration not fairness
Search/IR	BM25 0.0133 > dense 0.0067	V2 — seed-item NDCG/MRR
Two-tower (ID+content)	R@20 0.0642 (lost to ALS)	V3 G32 — ALS-floor protocol
Two-tower content ablation	+0.021 R@20 from MiniLM	V3 G32
Thompson vs greedy	cov 17→51%; Gini 0.97→0.81; rel -1.3pp	V3 G33 — offline sim
OPE of exploration policy	ESS 168→21; DR fails bandit	V3 G33 — methodology demo

Model registry

Key	Lane	Headline	Comparability
d3aplust_als_f64	V1	R@20 0.0846	V1 protocol ONLY
sasrec_canonical_v3	V1	R@20 0.065 (plot_reconstructed)	V1; packaging-limited
c2_als_f64	V2 SERVED	R@20 ~0.0375	V2 protocol ONLY
minilm_content	V2	reach 54.6%	candidate-gen only
g28_lambdamart_fusion	V2	test 0.036 (81 pos)	offline test split only
g32_two_tower	V3	R@20 0.0642 (honest negative)	ALS-floor protocol
g33_thompson_bandit	V3	cov 51%; Gini 0.81 (offline)	exploration sim, proxy reward

7 Gate-by-Gate Walkthrough (V2-V3)

Gate	What shipped	Result / verdict
G22	FAISS latency-quality frontier	FlatIP default lossless –47% p95; HNSW overlap 0.992; IVF rejected
G23	Production-shaped FastAPI + load test	warm p95 ~3.3ms; 0% empty; failure modes graceful
G24	Cold-start / sparse-cohort fallback quality	coverage collapse ~59×; recall non-monotonic; 57% item-cold-start
G25	Catalog exposure governance	all Gini>0.97; popularity degenerate; HNSW exposure-neutral
G26A	Decision-architecture spine + claim ladder	model/protocol registry; OPE logging schema designed
G26B	Heuristic exposure rerank, in-service	head-cap: coverage 6.9→9.8%, tail ~4.3×, no relevance loss; default OFF
G27	Semantic content retrieval (MiniLM+FAISS)	reach 54.6% of unseen golds; coverage 62.6% vs 8.1%; loses head
G28	Learned fusion tournament (LambdaMART)	beat ALS-only on test (0.036 vs 0.022) + recovered cold-start
G29	Off-policy evaluation executed	IPS/SNIPS/DR recover known value; DR lowest-bias, SNIPS stable
G30	Final RiskFrame audit + interview kit	9.1/10; offline gold-complete
G31	Search/IR front end (BM25+dense+RRF)	BM25 strongest single lane; NDCG/MRR; role coverage added
G32	Two-tower neural retrieval (ID+content, BPR)	R@20 0.064 vs ALS 0.085 — honest negative; content ablation +0.021
G33	Thompson Sampling exploration bandit	coverage 17→51%, Gini 0.97→0.81, –1.3pp relevance; OPE overlap collapse

The discipline an interviewer should notice: every gate has a pre-registered question, an artifact, and an honest verdict — including the gates that produced negatives (G20 SASRec, G22 IVF) and the gate that refused an expected story (G24).

V3 Neural Two-Tower + Thompson Exploration Bandit

V3 answers the two questions the target role weighs most heavily: can a neural two-tower with content beat ALS, and does a bandit exploration layer buy real catalog diversity? The answers are an honest **no** and a measured **yes-with-a-cost** — and the sharpest result is that off-policy evaluation breaks down for the exploration policy exactly where the theory says it should.

G32 — two-tower vs ALS (identical protocol)

User tower: mean-pooled learned embeddings of the user's history (≤ 50 items). Item tower: a learned item-ID embedding plus a projection of the 384-d MiniLM content embedding. BPR loss, 8 uniform negatives, 23 epochs. Evaluated leave-last-out on the same 5,000-user eval_sample, top-200, history-masked, with per-user bootstrap 95% CIs — the exact ALS-floor protocol.

Model	Recall@20	95% CI	NDCG@20	Verdict
ALS (F=64)	0.0846	[0.0766, 0.0922]	0.0344	baseline
Two-tower (ID + content)	0.0642	[0.0574, 0.0712]	0.0262	loses to ALS
Two-tower (ID only, content off)	0.0432	[0.0376, 0.0490]	0.0162	ablation

- **ALS wins, not noise:** its CI lower bound 0.0766 is above the two-tower's upper bound 0.0712 — non-overlapping.
- **Content is a real signal:** the content tower adds +0.021 R@20 (0.0432 → 0.0642), CIs non-overlapping — content is now first-class inside retrieval.
- **Second neural negative:** SASRec (0.065) and the two-tower (0.064) both lose to MF on dense warm book histories — depth is not the lever.

G33 — Thompson Sampling exploration (offline)

For each of the 5,000 warm users, take the two-tower top-200 pool and build a size-20 slate under greedy (rank by score) vs Thompson (per-item Beta posterior, updated with a held-out-positive proxy reward over 6 passes). Offline simulation — no live serving.

Metric	Greedy	Thompson	Direction
Catalog coverage	16.6% (6,972)	50.5% (21,201)	~3× more catalog exposed
Exposure Gini	0.9736	0.8063	less concentrated
Intra-list diversity	0.492	0.5054	modestly higher
Slate hit-rate@20 (relevance)	0.0642	0.0510	−1.3pp (the cost)

Off-policy evaluation of the two policies (IPS/SNIPS/DR, single-action bandit)

Target	true	IPS	SNIPS	DR	ESS	matches
Greedy top-1	0.0062	0.0048	0.0047	0.0050	168	332
Thompson top-1	(0.018, biased)	0.0000	0.0000	−0.0005	21	38

HONESTY BOUNDARY (V3)

Offline only. The two-tower LOST to ALS (reported as an honest negative). The bandit REDUCES top-slate relevance — it buys exploration/diversity, not accuracy. OPE recovers the greedy policy but is unreliable for the exploration policy (ESS 168→21), so its estimates are a methodology demo, not online value. The Thompson 'true value' is optimistically biased (posterior fit on the same proxy reward) and is not counted as a relevance gain. No online lift, no A/B, no production claim.

8 Off-Policy Evaluation — deep dive

Setup. Single-action contextual-bandit OPE. Context x = user; action a = one item drawn from the candidate pool; the stochastic logging policy $\pi_0 = \epsilon$ -uniform ($\epsilon=0.15$) + softmax over an inverse-rank blend, guaranteeing overlap ($\pi_0>0$). Reward $r = 1$ if the drawn item is in the user's held-out test interactions. Evaluated on the leakage-safe TEST users; 30 Monte-Carlo draws/user $\rightarrow$ 28,320 logged samples.

Estimators

- **IPS** = $(1/N) \sum [1\{a_i=\pi_e(x_i)\} / \pi_0(a_i|x_i)] r_i$ — unbiased, high variance.
- **SNIPS** = $\sum w_i r_i / \sum w_i$ — self-normalized; lower variance, slight bias.
- **DR** = $\text{mean}(\hat{r}(x, \pi_e)) + \text{mean}(w_i(r_i - \hat{r}(x, a_i)))$ — lowest bias with a calibrated reward model $\hat{r}$.

Result (recover a known value)

Target policy	true (precision@1)	IPS	SNIPS	DR	ESS
ALS top-1	0.0106	0.0211	0.0129	0.0089	266
RRF-fusion top-1	0.0117	0.0218	0.0134	0.0110	220

Reading: all three land near the directly-computed true value — the logged-propensity pipeline is correct. DR is closest (its reward-model baseline reduces bias); IPS over-estimates $\sim 2\times$ (the textbook high-variance failure at modest ESS); SNIPS sits between. The estimators correctly rank RRF-fusion above ALS — the ordering OPE would use to choose a policy.

HONESTY BOUNDARY

Propensities are synthetic (a logging policy we control) and the reward is a held-out-gold proxy, not real engagement. This VALIDATES the estimators offline; it does NOT estimate real online value. A real off-policy number needs logged real-traffic propensities — the labelled next step.

9 Cold-Start & Exposure — deep dive

Cold-start taxonomy

User cold-start: unknown (0 history) / sparse (1-2) / low (3-5) / warm (6+). **Item cold-start (offline-split sense):** 57.2% of held-out gold items are unseen in train — no collaborative signal, so ALS/popularity reach them with probability ~ 0 by construction.

The honest finding (not the expected one)

Per-cohort Recall@20 is **non-monotonic** — cold/sparse cohorts score equal-or-higher than warm, because a low-history user's next item is more popularity-predictable and easier to retrieve. So **recall understates the cold-start cost**. The real, measured cost of falling back to popularity is a **collapse of personalization and catalog coverage**:

Metric (top-20)	ALS warm (served)	Popularity fallback
Unique items recommended	3,532	60
Catalog coverage	8.7%	0.15%
Slate personalization (distinct/users)	0.68	0.001
Long-tail exposure share	0.4%	0.0%

Exposure governance

Over the full catalog (zeros included): every policy is concentrated (Gini > 0.97); popularity is near-degenerate (Gini 0.9995, 99.9% zero-exposure, ~ 60 unique items). Head items get $\sim 10\times$ their catalog base rate; the long tail gets $\sim 0.02\times$ (suppressed $\sim 50\times$). HNSW is exposure-neutral vs exact (Gini $0.986 \approx 0.986$). This is catalog-exposure **concentration** governance — explicitly NOT protected-class fairness.

Mitigation (G26): a head-item exposure cap is Pareto-dominant on the warm cohort — it improves Recall@20 +40% AND long-tail exposure $\sim 8\times$ AND lowers Gini, because head inflation was crowding niche relevant items out of the top-20. MMR and genre-diversification were honest negatives. The win is protocol-specific (warm cohort, single-held-out-gold).

10 Truth Boundary & Claim Ladder

Real (built & measured)	Simulated / proxy	NOT claimed
2.56M Goodreads dataset, temporal split	OPE propensities synthetic	online lift / engagement / A/B impact
ALS/SASRec/content/semantic/BM25 retrieval	OPE reward = held-out-gold proxy	production quality (deployed $\neq$ effective)
FAISS serving + load test + Cloud Run deploy	81 test positives in fusion eval	solved cold-start / long-tail
cohort recall, exposure Gini, NDCG/MRR, IPS/SNIPS/DR	—	fairness certification
gate-by-gate audit history	—	LLM recommender / semantic taste
model/protocol registry	—	learned-fusion-beats-ALS beyond the test split

Claim ladder

- **Locked safe:** offline decision dossier · ALS warm floor · FAISS frontier · fallback quality · exposure governance · deployed-to-Cloud-Run.
- **Safe with boundary:** SASRec converged-and-lost (plot_reconstructed) · production-LIKE API · cold-start handled at system level · OPE methodology demonstrated.
- **Forbidden:** online lift · real users · real off-policy value · cold-start solved · fairness certified · LLM recommender · learned-beats-ALS in production.

11 Product & Business Reasoning

Decision	First-principles driver	Business consequence
ALS = warm floor	depth didn't beat MF at convergence	ship a simple, cheap, strong floor
coverage > sophistication	recall ceiling is candidate-bound	invest in candidate generation, not depth
FAISS = latency evidence	ANN doesn't change relevance	serving cost, not a UX lift
fallback existence $\neq$ quality	always-nonempty $\neq$ good	cold UX is generic; measure it
exposure audit $\neq$ fairness	concentration $\neq$ protected-class	catalog-health signal, not certification
popularity = emergency only	Gini 0.9995, zero personalization	creator/tail starvation if it dominates
logs for real OPE	IPS needs $\pi_{\log}$ overlap	online decisioning unlocked once logged

Who uses this: recommender DS (protocol/metric validity), ranking ML engineer (retrieval/serving tradeoffs), search/discovery & marketplace PM (catalog health, cold-start UX), platform engineer (latency, fallback, monitoring). **Why wrongness is expensive:** a recall-looking win that worsens catalog health, an ANN setting that shreds the candidate pool, or a popularity fallback that quietly becomes dominant all degrade discovery and creator opportunity at scale before any dashboard notices.

12 Hard Q&A

Q. Why did SASRec lose to ALS?

A. Book next-item is weakly sequence-predictable; the dominant signal is collaborative co-read, which MF captures directly. I trained a canonical full-softmax SASRec to convergence and it still scored below ALS — depth was not the lever on sparse histories.

Q. Was the SASRec undertrained?

A. No — I monitored convergence on eval-R@20 after catching a NaN val-loss false-stop, and showed the full-softmax-vs-sampled lever ($\sim 6.7\times$) was the real driver; even fixed, it didn't beat ALS.

Q. What's the OPE variance problem and how does SNIPS fix it?

A. Raw IPS weights π_e/π_0 blow up variance at modest ESS (my IPS over-estimated $\sim 2\times$). SNIPS self-normalizes by the sum of weights, trading a little bias for much lower variance — my recommended estimator when no reward model exists; DR is lowest-bias when a calibrated one does.

Q. Why did DR go negative once?

A. A class-balanced reward model calibrated toward ~ 0.5 vs a true $\sim 1\%$ base rate, so it over-predicted and broke the DR baseline. I caught it from the impossible negative value and switched to a base-rate-calibrated model; DR then recovered the known value best of the three.

Q. What's the real cost of cold-start here?

A. Not a recall drop — recall is non-monotonic because cold users' next item is more popularity-predictable. The real cost is coverage collapse $\sim 59\times$ and personalization $0.68 \rightarrow 0.001$. I reframed the metric around what actually degrades.

Q. Why reject IVF if it was faster?

A. Low-nprobe IVF kept gold Recall@20 flat while candidate-overlap vs exact collapsed to 0.28–0.78 — a broken pool a single-gold metric can't see. Speed that destroys candidate fidelity is a recall bug.

Q. Why did BM25 beat dense on short text?

A. On title+genre text, exact lexical overlap (shared series/genre words) carries more signal than a 384-d embedding that blurs the rare disambiguating tokens. Embeddings win on long paraphrase-heavy text; lexical wins on short keyword-dense text.

Q. So is your hybrid retrieval better?

A. On recall, no — RRF hybrid sits below BM25 alone on this seed-item task. Its value is robustness: it never collapses to either lane's weakness and it reaches cold-start. I don't claim hybrid beats both.

Q. Did the learned ranker actually beat ALS?

A. On the held-out test split, yes (0.036 vs 0.022) and it recovered cold-start (0.032 vs 0), consistent across four model families — but on 81 positives, single-gold, offline. Directional, not a production claim.

Q. Why does fusion help at all?

A. ALS and semantic candidate sets are almost disjoint (overlap 6,749 of $\sim 880k$). A learned ranker picks the best across both complementary sources — that's what lifted test recall and cold-start.

Q. What does the live Cloud Run endpoint serve?

A. The served-c2 ALS model with exact FlatIP retrieval: /recommend returns top-K with source tags and a graceful fallback (unknown user $\rightarrow$ popularity, never empty). /health reports readiness. "Deployed and serving HTTPS" is true; "improved engagement" is not.

Q. Is any of this production / online?

A. No — offline by scope. The API is deployed and callable, but there are no real users, no logged real-traffic propensities, and no A/B. That boundary is documented as the next step.

Q. Why is your served recall ~0.0375 when V1 was 0.0846?

A. Different builds and protocols — d3aplustuning vs served-c2 full-catalog single-held-out-gold. I keep them in a registry so they're never conflated. The relative cross-policy structure is the V2 signal, not the absolute.

Q. Is the semantic lane an LLM recommender?

A. No — embedding-based candidate generation (MiniLM over content text). No generation, no chat, no "semantic taste." It reaches items collaborative filtering can't.

Q. Did the V3 two-tower beat ALS?

A. No — and I report it as an honest negative. A proper two-tower (mean-pooled history user tower + learned-ID@MiniLM-content item tower, BPR) reached R@20 0.0642 [0.057-0.071] vs ALS 0.0846 on the identical protocol; the CIs don't overlap. The useful result is the content ablation: MiniLM content adds +0.021 R@20, so content earns its place inside retrieval even though the model still loses to MF on dense warm histories. Two neural models (SASRec, two-tower) now lose to ALS — consistent, not a fluke.

Q. What did the Thompson Sampling bandit actually buy?

A. Exploration, at an explicit relevance cost — not a relevance win. Over two-tower candidate pools, Thompson vs greedy raised catalog coverage 16.6%→50.5% and cut exposure Gini 0.97→0.81, for a ~1.3pp drop in top-slate hit-rate (0.064→0.051). That's the explore/exploit tradeoff, stated as a cost. Offline simulation with a proxy reward — no online lift.

Q. Why did OPE break for the bandit but not for greedy?

A. Overlap. The synthetic logging policy covers the greedy top-1 well (332 matched samples, ESS 168) so IPS/SNIPS/DR recover it (DR 0.005 vs true 0.006). The exploration policy selects items the logging policy rarely logs, so overlap collapses (38 matches, ESS 21) and the estimators become unreliable — DR even goes negative. You can't off-policy-evaluate an exploration policy a logging policy doesn't cover; it's the same lesson as the G29 DR-negative failure.

Q. Is your exposure work a fairness claim?

A. No — catalog exposure concentration (Gini/coverage/tier-lift), explicitly not protected-class fairness. I don't certify fairness; I measure concentration as a ship-gate metric.

Q. How did you prevent leakage in the fusion eval?

A. Split by user-hash (a user is wholly in one split); test labels never used for training or model selection; served-c2 protocol only. The fix for the v2 SASRec leakage (stopping on the eval set) was a separate validation-monitor set.

Q. Why is the cover claim 'gold_candidate' not 'gold_complete'?

A. One packaging gap: the G20 SASRec headline is plot_reconstructed (the direct artifact was lost). The decision is sound; the evidence packaging is one artifact short, so I hold the status honestly at candidate.

Q. What would a head-cap rerank cost in production?

A. Sub-millisecond, and it improved recall AND exposure on the warm cohort here — but the win is protocol-specific (single-held-out-gold). I ship it config-flagged, default OFF, and would tune the cap on a live relevance target.

Q. What's your single biggest risk if I pushed you?

A. Small-sample fragility — the fusion win rests on 81 test positives. I state it, it's directionally consistent across four model families, and the honest resolution is propensity logging + an A/B, which I've designed but not run.

Q. Could you have just shipped the best offline number?

A. That's exactly the failure mode the project targets. The IVF and head-cap stories show a better-looking number that was either broken (IVF) or protocol-specific (head-cap). Shipping a metric without its mechanism is how recsys quietly breaks.

Q. What would falsify your headline claims?

A. ALS-floor: a properly-tuned sequence/graph model beating it on this exact protocol. Fusion-win: not holding on a larger positive set. Semantic value: cold-start reach not translating to online engagement. OPE: ESS too low / no overlap. I state these kill conditions, not hide them.

13 What I'd Build Next

With production access, in priority order:

- **Real-traffic propensity logging** — populate the designed OPE schema (request/impression/position/ $\pi_{\log}$ /reward) to turn IPS/SNIPS/DR into real off-policy value estimates.
- **Canary + live A/B** of the fusion policy and the head-cap exposure rerank, with guardrail metrics (relevance, coverage, Gini, fallback rate, p95) wired to alerts — the only way to claim online lift.
- **In-ranker position-bias correction (PAL / IPS-weighted LTR)** — upgrade position bias from *evaluated* to *corrected*.
- **Productionise the lexical lane** (Lucene / Elasticsearch / bm25s) for sub-ms BM25, and persist the FAISS index to cut cold-start build time.
- **Two-tower & exploration are now built (V3)** — the two-tower is an honest negative with a real content lift, and the Thompson bandit needs real logged propensities to make its OPE trustworthy; graded-relevance labels and GNN retrieval remain the next scale-up.

14 Evidence Artifact Index

Every claim maps to a JSON in `outputs/evidence/`. Selected artifacts:

Artifact	Proves
<code>g22_faiss_latency_quality_report.json</code>	FlatIP lossless –47% p95; HNSW near-lossless; IVF overlap collapse
<code>g23_serving_api_report.json</code>	production-like API, 0% empty, failure-mode coverage
<code>g24_cold_start_sparse_cohort_report.json</code>	fallback coverage collapse ~59×, personalization 0.68→0.001
<code>g25_catalog_exposure_governance_report.json</code>	exposure Gini/coverage; HNSW exposure-neutral
<code>g26b_rerank_integration_report.json</code>	head-cap rerank: +recall, +coverage, no relevance loss
<code>g27_semantic_retrieval_report.json</code>	semantic reaches 54.6% of item-cold-start golds
<code>g28_final_ranker_fusion_decision.json</code>	learned fusion beats ALS-only on held-out test
<code>g29_ope_execution_report.json</code>	IPS/SNIPS/DR recover a known value offline
<code>g31_search_ir_report.json</code>	BM25 + dense + RRF with NDCG/MRR
<code>g30_final_gold_audit.json</code>	RiskFrame 9.1, offline gold-complete
<code>g_serving_stress_test.json</code>	serving edge-case stress test + two load-path fixes
<code>G32_two_tower_eval.json</code>	two-tower R@20 0.064 vs ALS 0.085 + content ablation (V3)
<code>G33_bandit_eval.json</code>	Thompson coverage/Gini/diversity + OPE overlap collapse (V3)

Companion docs in the repo: `PULSEDISCOVERY_INTERVIEW_KIT.md` (pitch & claim ladder) ·
`PULSEDISCOVERY_UNIFIED_DEFENSE_KERNEL.md` (method-by-method defense) ·
`PULSEDISCOVERY_FAILURES_AND_HARDENING.md` (failures + serving stress test) ·
`G26A_MODEL_PROTOCOL_REGISTRY.md` (V1/V2 registry).

If a result isn't backed by an artifact in `outputs/evidence/`, it isn't claimed.